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Goal

- Gemini targets joint understanding of language, images, audio, video and code instead of attaching isolated modality tools.
- The family spans difficult reasoning services and memory-constrained on-device applications.

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Language reasoning

- Gemini Ultra reaches 90.0% on MMLU with uncertainty-routed chain-of-thought, above the reported human-expert baseline.
- Across the report it advances the state of the art on 30 of 32 broad benchmarks.

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Three scales

ULTRA, PRO AND NANO

- Ultra is the most capable model, Pro balances quality and serving cost, and Nano targets mobile deployment.
- Models are trained on multimodal and multilingual data and optimized on Google's TPU infrastructure.

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Multimodal evidence

STATE OF THE ART ON 20 OF 20

- The report records improvements on all 20 tested multimodal benchmarks covering image, video and audio understanding.
- Cross-modal reasoning includes chart reading, document understanding, speech, video and code generation.

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Training stack

PRETRAIN, INSTRUCT, ALIGN

- Native multimodal pretraining exposes the model to interleaved modalities and code before task adaptation.
- Instruction tuning and reinforcement learning from human feedback improve helpfulness and dialogue behavior.

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Limits & responsibility

- Results depend on prompting and evaluation protocols; model architecture, data and contamination controls are only partially described.
- Bias, hallucination and multimodal misuse remain, motivating safety evaluations, filtering and staged deployment.

GEMINI 1.0

Gemini: A Multimodal Model Family

Gemini Team | arXiv:2312.11805v5

Build one natively multimodal model family across text, images, audio, video and code, spanning on-device efficiency to frontier reasoning.

native multimodality

Ultra Pro Nano

broad evaluation